

Scalable AI

Introduction to HPC and Slurm



What is **slurm** ?
workload manager



Resource Manager



Job Scheduler

Job ID	Job Name	State	Nodes	Time
123456789	test-job	Pending	1	00:00
987654321	test-job	Running	1	00:01
111111111	test-job	Completed	1	00:02
222222222	test-job	Failed	1	00:03

Basic interaction

What is Slurm?



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Cluster manager and job scheduler
system for large and small Linux
clusters.

- **Simple Linux Utility for Resource Management** (Historic)
or simply **Slurm**.



What is Slurm?



Cluster manager and job scheduler system for large and small Linux clusters.

- Allocates access to resources for some duration of time.
- Provides a framework for starting, executing, and monitoring work (normally a parallel job).
- Arbitrates contention for resources by managing a queue of pending work.

Resource Managers

ALPS (Cray)

Torque

Scheduler

Maui

Moab

LoadLeveler (IBM)

LSF

Slurm

PBS Pro

What is Slurm?

Resource Managers

ALPS (Cray)

Torque

Scheduler

Maui

Moab

- ✓ Open source
- ✓ Fault-tolerant
- ✓ Highly scalable
- ✓ Almost everywhere

LoadLeveler (IBM)

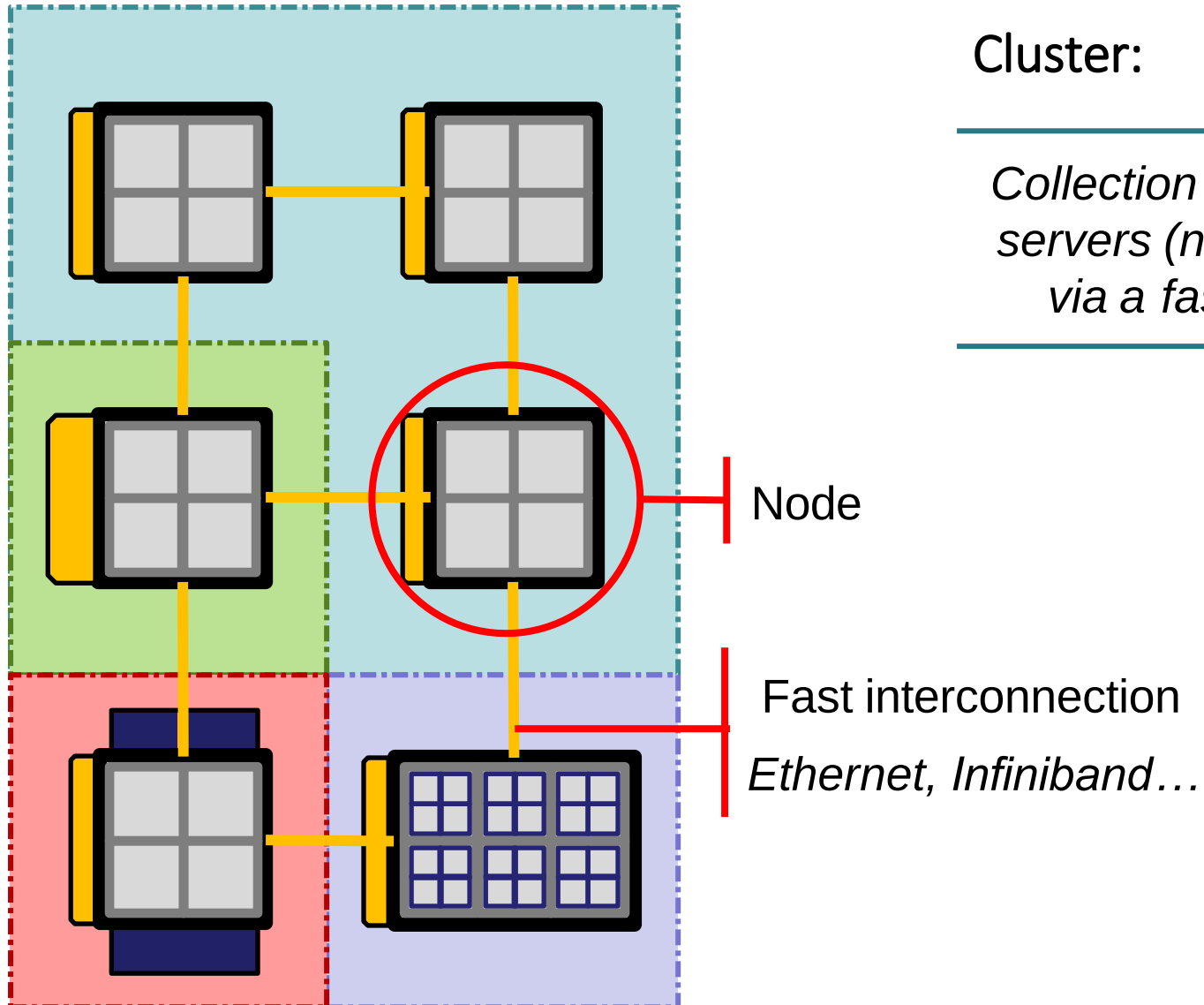
LSF

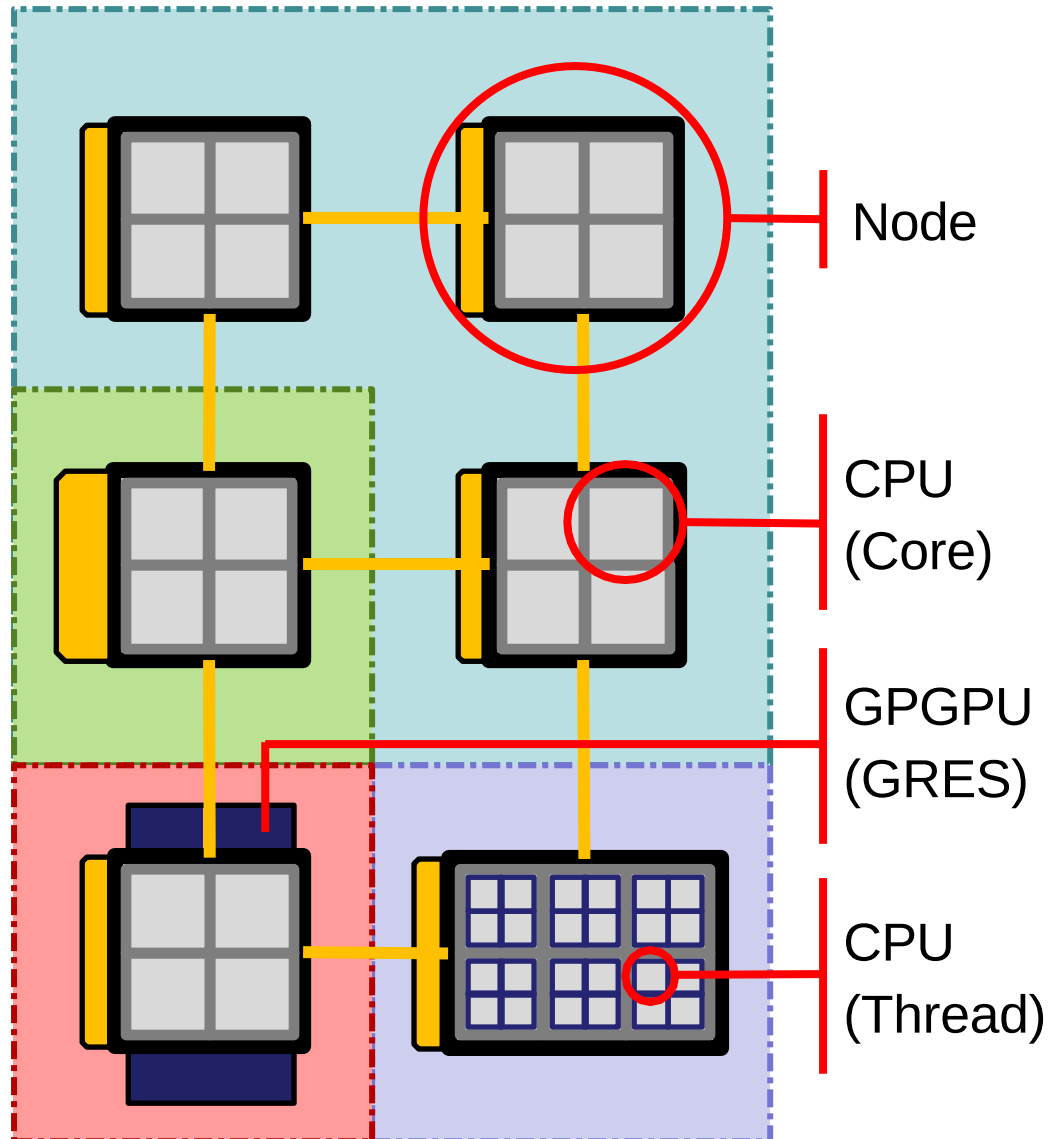
Slurm

PBS Pro

Cluster:

Collection of many separate servers (nodes), connected via a fast interconnect.

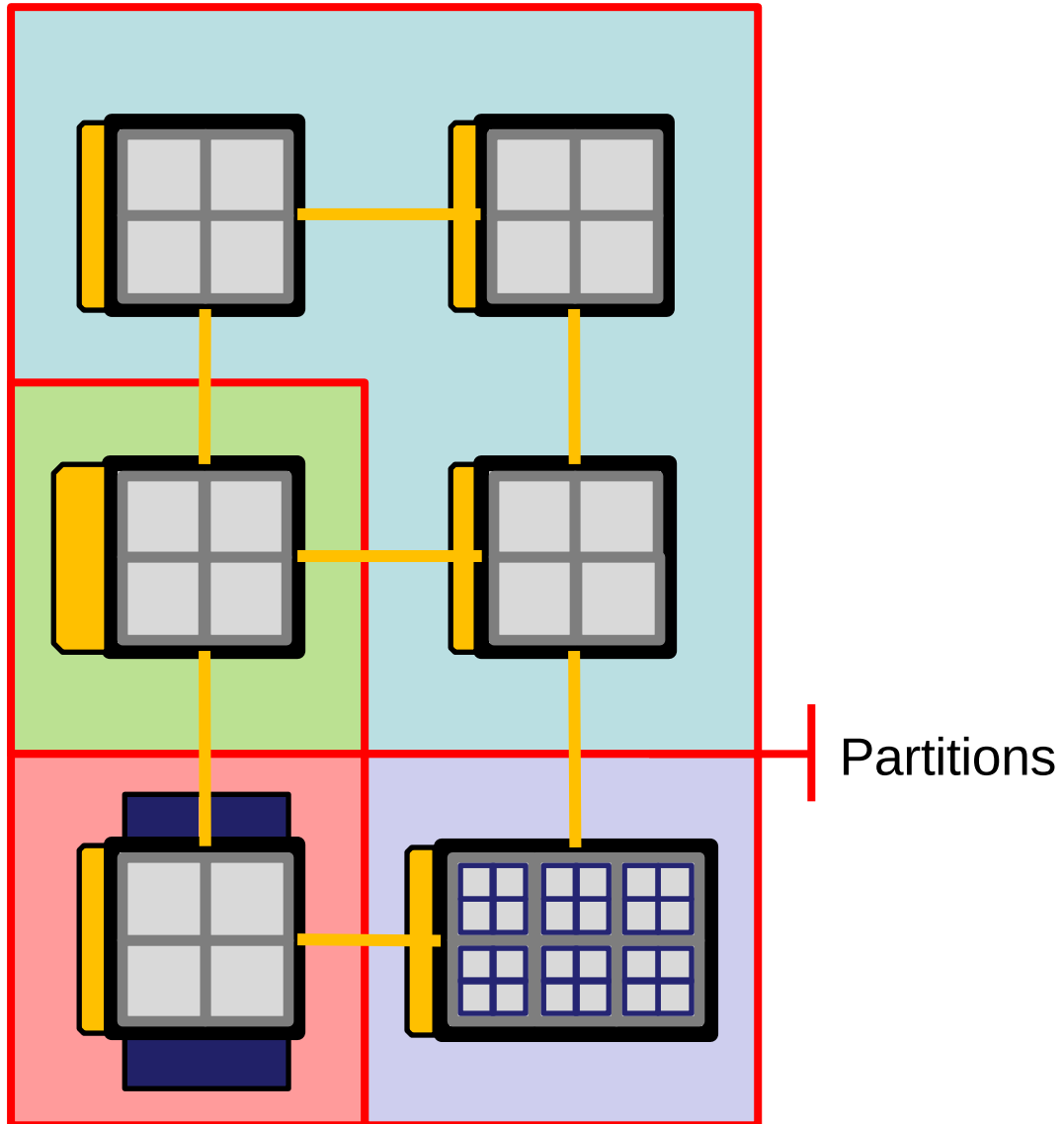




Nodes:

Individual computer component of an HPC system.

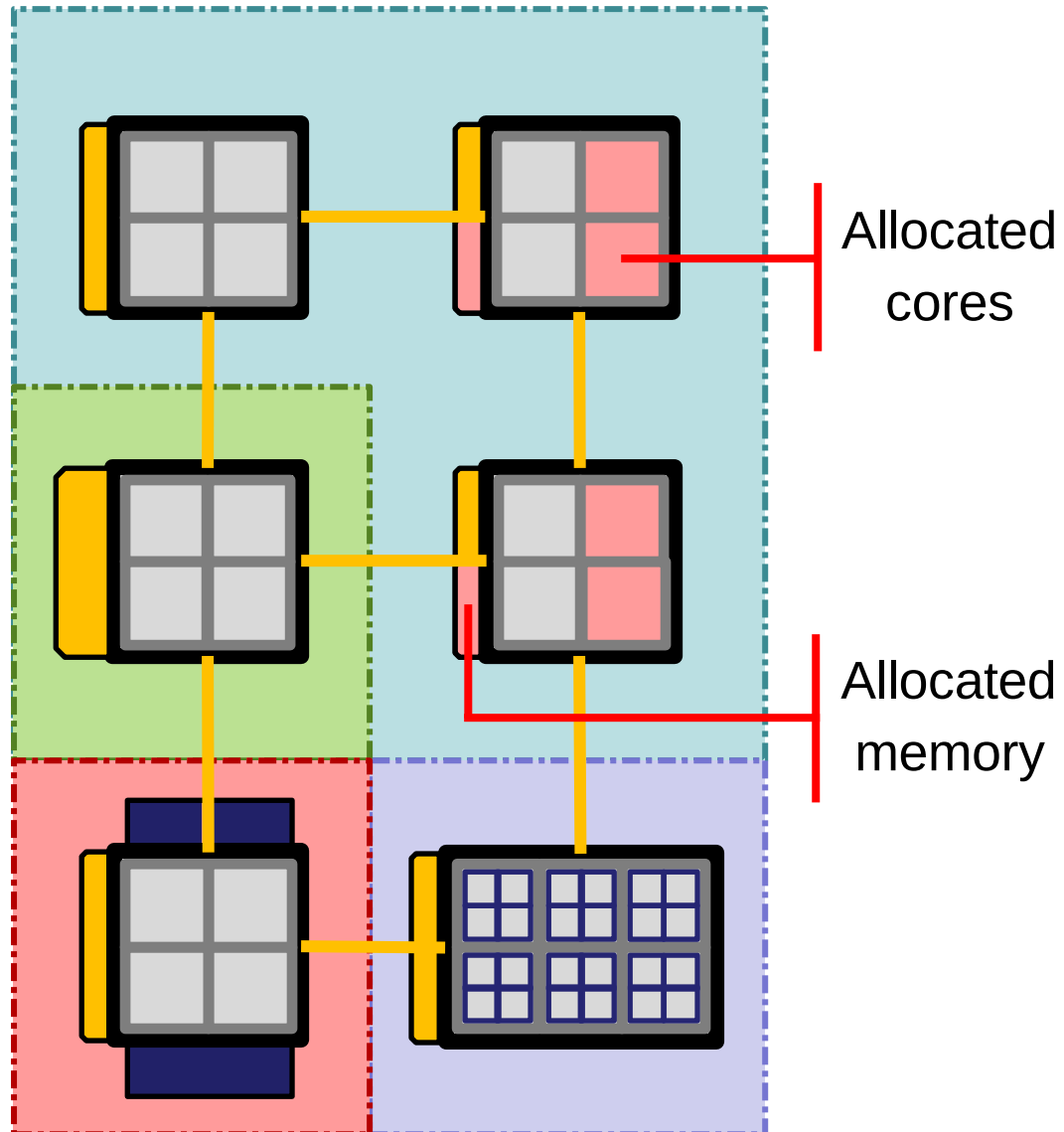
- `lrndn[0001-4988]`



Partitions:

*Logical group of nodes with
common specs.*

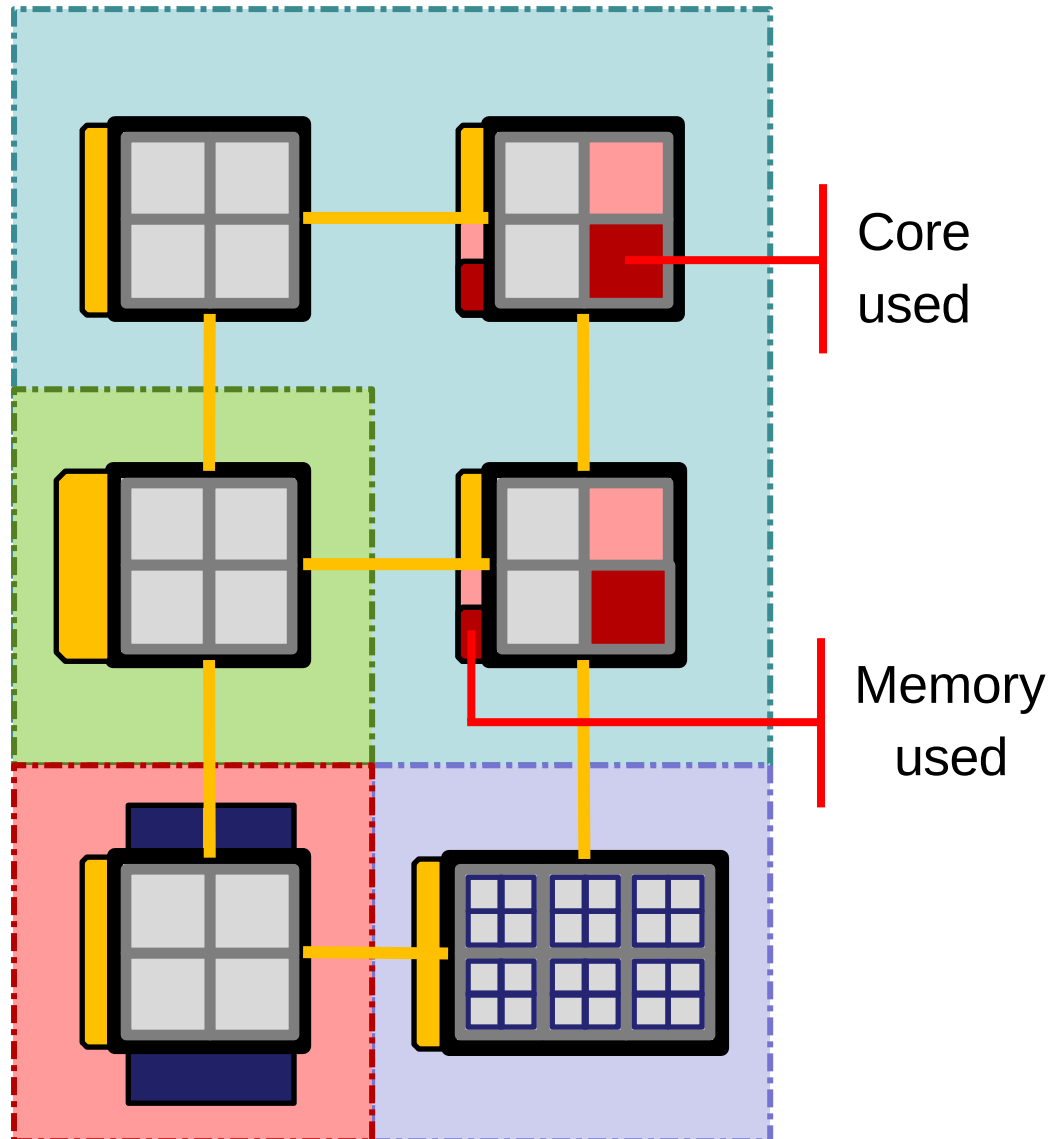
- lrd_all_serial
- boost_usr_prod
- dcgp_usr_prod
- boost_fua_prod
- boost_fua_dbg
- lrd_cin_viz



Jobs:

Allocations of resources assigned to a user for a specified amount of time.

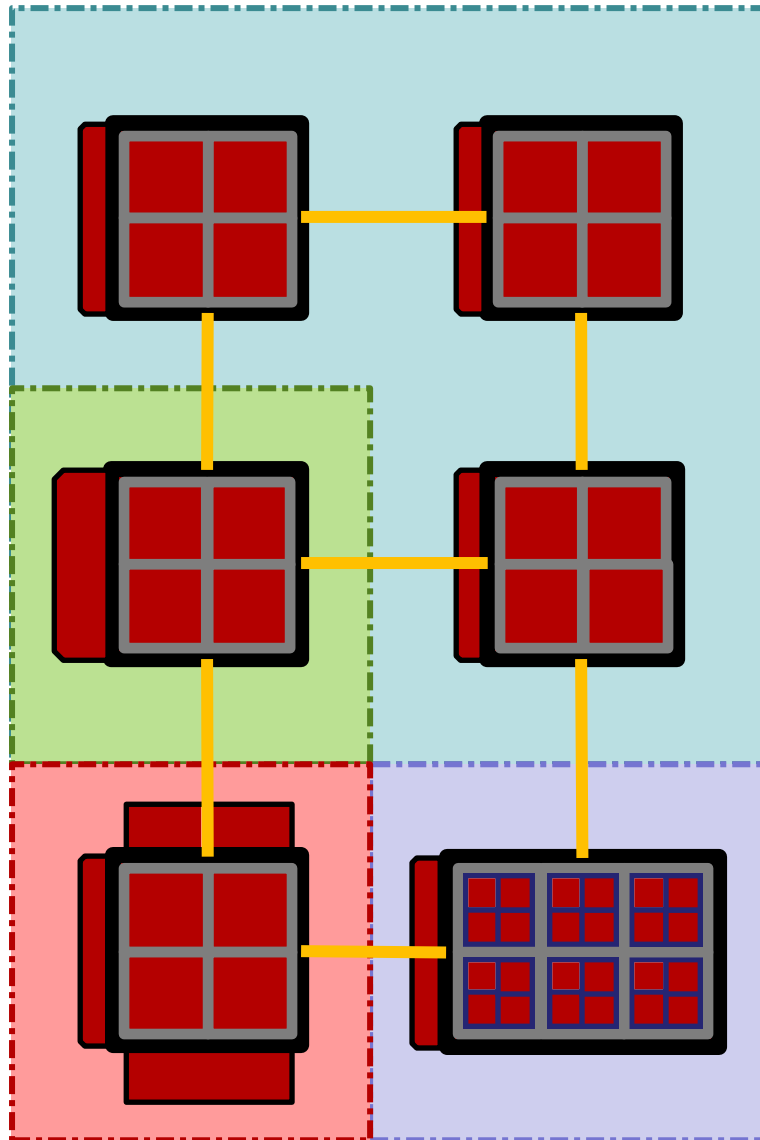
- ID (a number)
- Name
- Time limit
- Size specification
- Other Jobs Dependency
- State



Jobs Step:

Sets of (possibly parallel) tasks within a job.

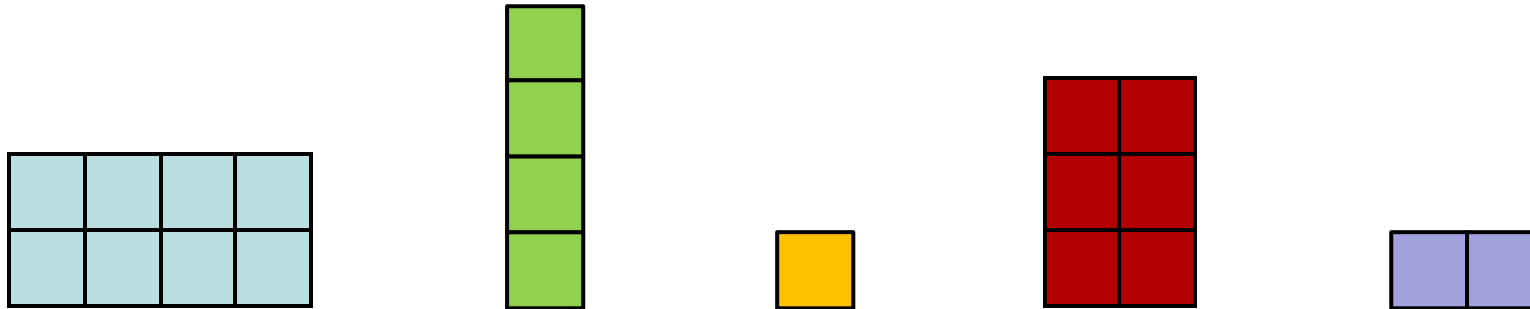
- ID (a number)
- Name
- Time limit
- Size specification



FULL CLUSTER

Job scheduling time!

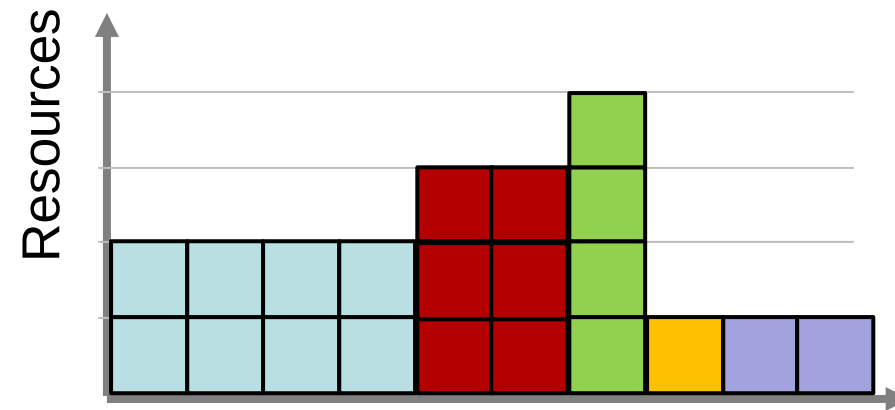
Scheduling: The process of determining next job to run and on which resources.



Slurm: Job Scheduling

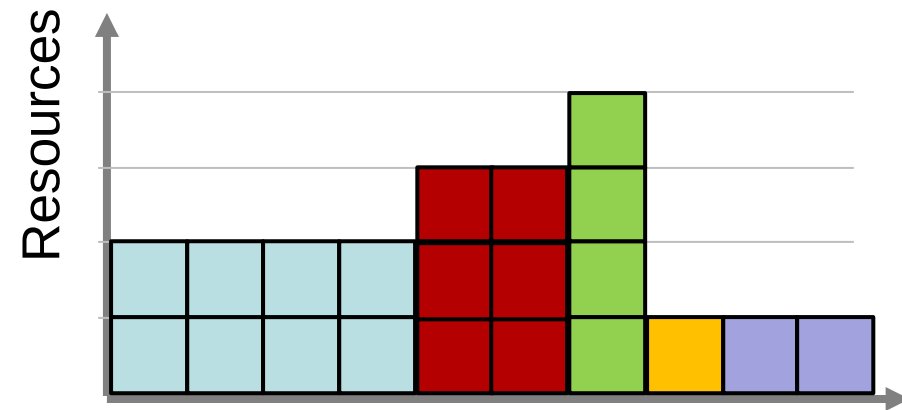
Scheduling: The process of determining next job to run and on which resources.

FIFO Scheduling



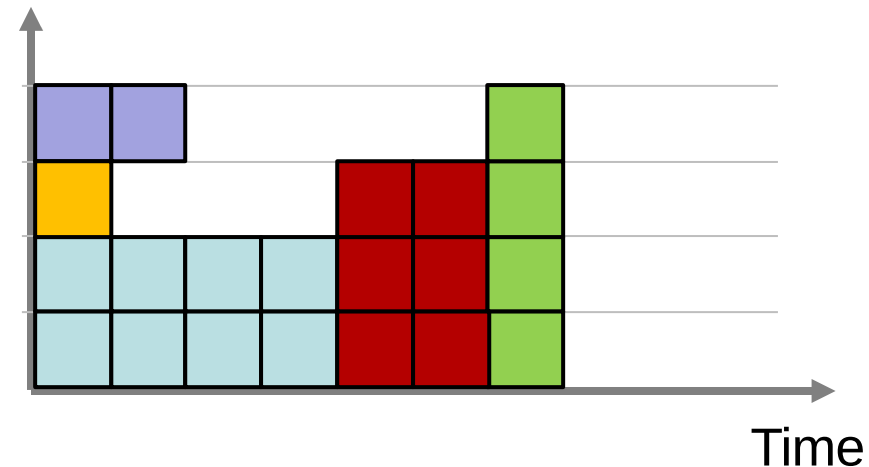
Scheduling: The process of determining next job to run and on which resources.

FIFO Scheduling



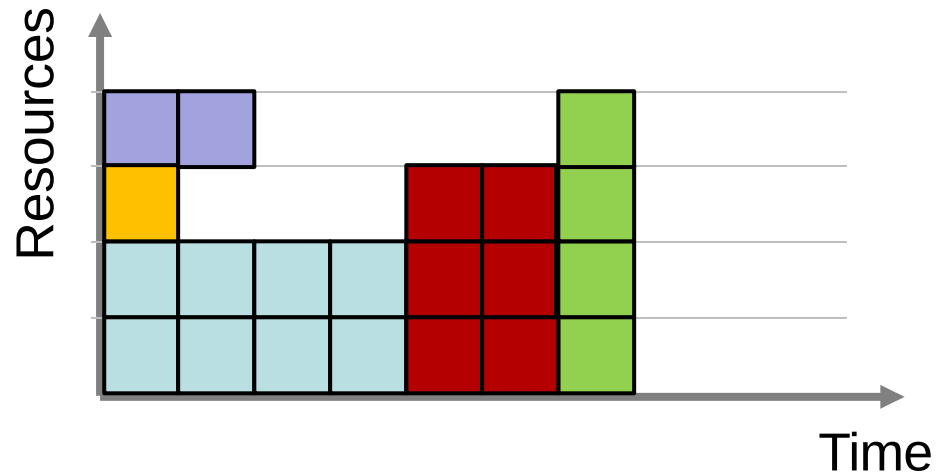
Backfill Scheduling

- Job priority
- Time limit (**Important!**)

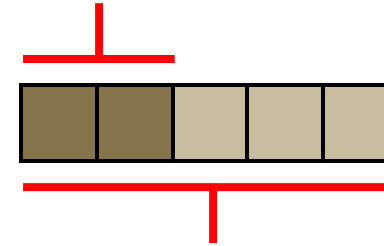


Backfill Scheduling:

- E_j : New lower priority job



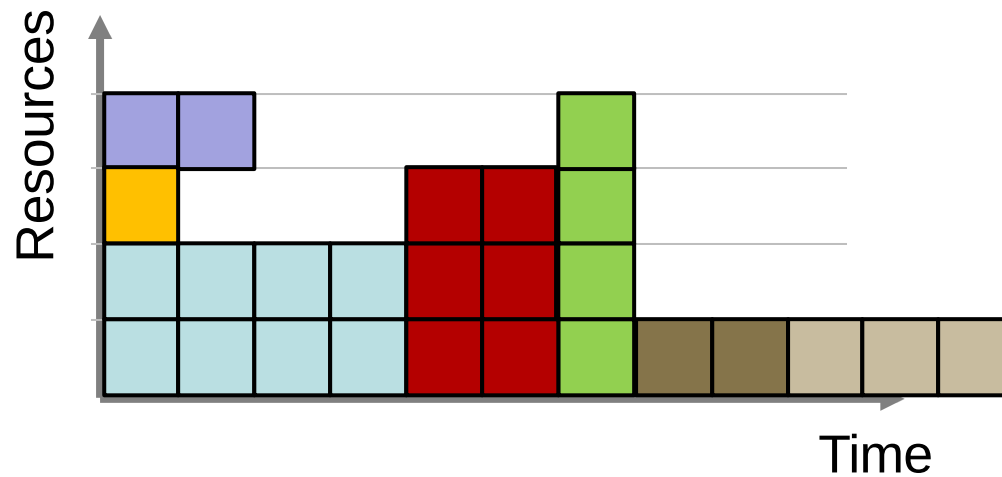
Elapsed time



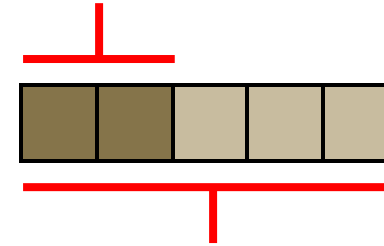
Time limit

Backfill Scheduling:

- Ej: New lower priority job



Elapsed time

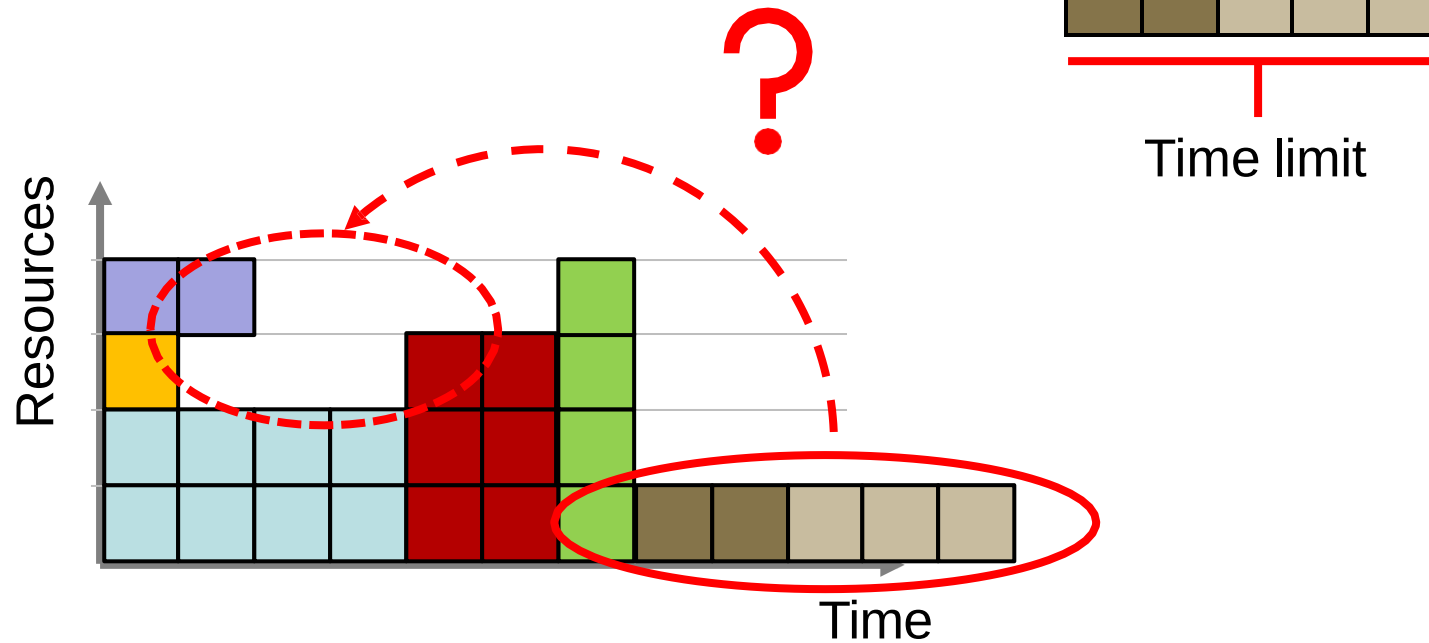


Time limit

Submit

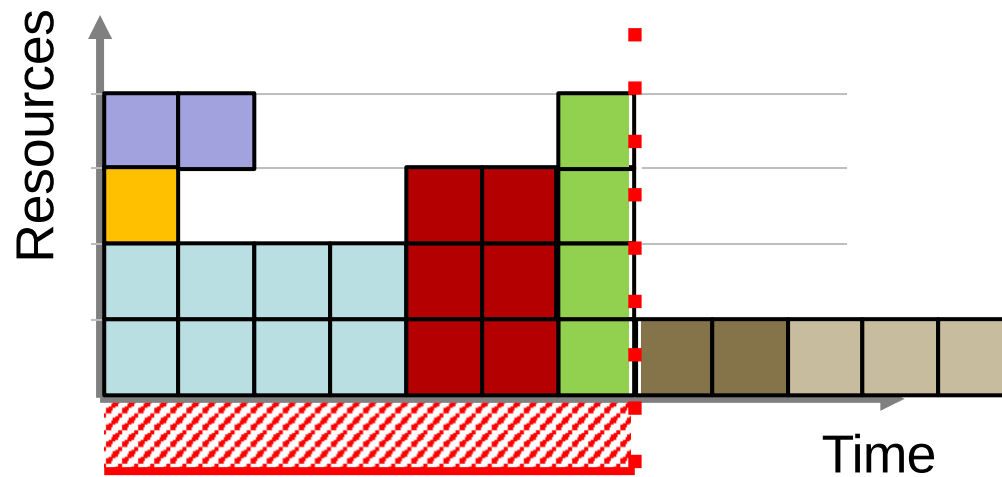
Backfill Scheduling:

- Ej: New lower priority job



Backfill Scheduling:

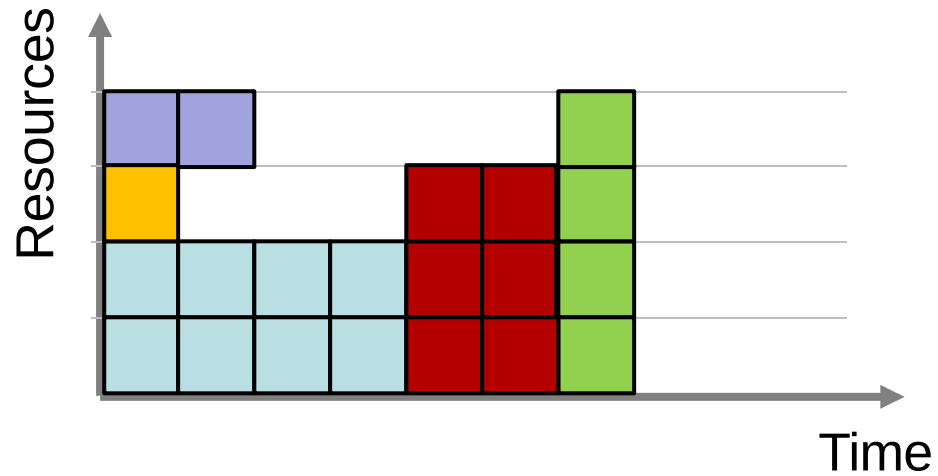
- Ej: New lower priority job



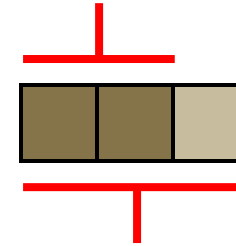
Wait time: 7

Backfill Scheduling:

- E_j : New lower priority job



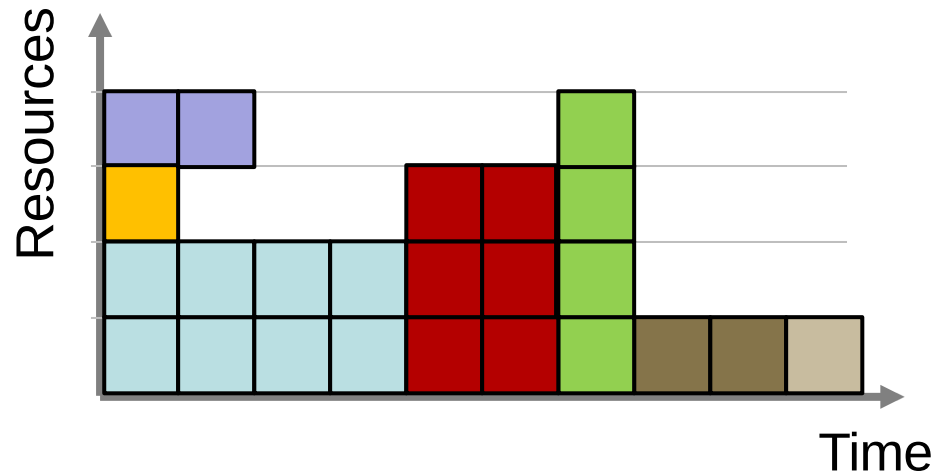
Elapsed time



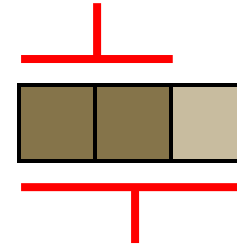
Time limit

Backfill Scheduling:

- E_j : New lower priority job



Elapsed time

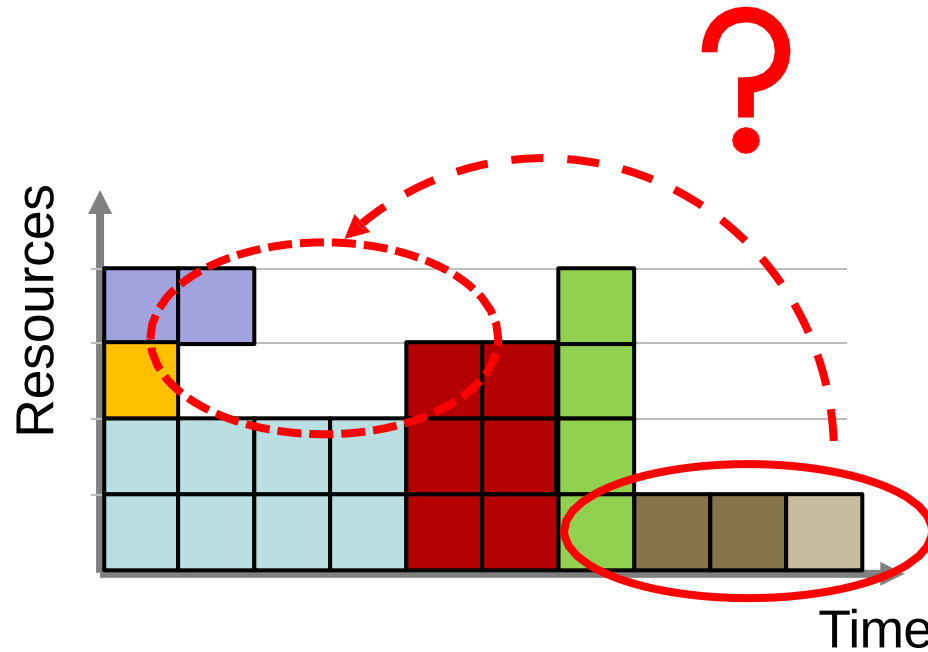


Time limit

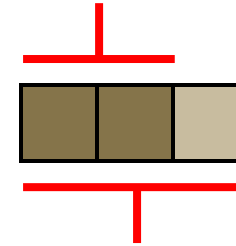
Submit

Backfill Scheduling:

- Ej: New lower priority job



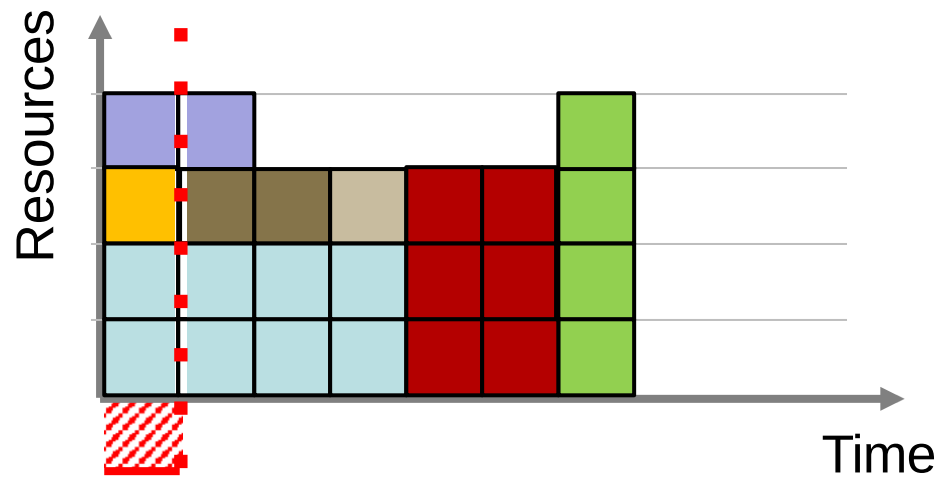
Elapsed time



Time limit

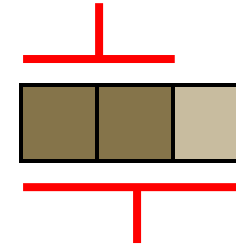
Backfill Scheduling:

- Ej: New lower priority job



Wait time: 1

Elapsed time



Time limit

Backfill Scheduling:

- Starts with job priority.

$$\begin{aligned} \text{Job_priority} = & \\ &= \text{site_factor} + \\ &+ (\text{PriorityWeightQOS}) * (\text{QOS_factor}) + \\ &+ (\text{PriorityWeightPartition}) * (\text{partition_factor}) + \\ &+ (\text{PriorityWeightFairshare}) * (\text{fair-share_factor}) + \\ &+ (\text{PriorityWeightAge}) * (\text{age_factor}) + \\ &+ (\text{PriorityWeightJobSize}) * (\text{job_size_factor}) + \\ &+ (\text{PriorityWeightAssoc}) * (\text{assoc_factor}) + \\ &+ \text{SUM}(\text{TRES_weight_}<\text{type}> * \text{TRES_factor_}<\text{type}>...) \\ &- \text{nice_factor} \end{aligned}$$

Backfill Scheduling:

- Starts with job priority.

Job_priority =

= site_factor +

+ (PriorityWeightQOS) * (QOS_factor) +

+ (PriorityWeightPartition) * (partition_factor) +

+ (PriorityWeightFairshare) * (fair-share_factor) +

+ (PriorityWeightAge) * (age_factor) +

+ (PriorityWeightJobSize) * (job_size_factor) +

+ (PriorityWeightAssoc) * (assoc_factor) +

+ SUM(TRES_weight_<type> * TRES_factor_<type>...)

- nice_factor

Fixed value

Dynamic value

User defined value

Backfill Scheduling:

- Priority factor:

QoS:

- Account's Priority:
 - Normal
 - Low
- RES users:
 - class_a
 - class_b
 - class_c



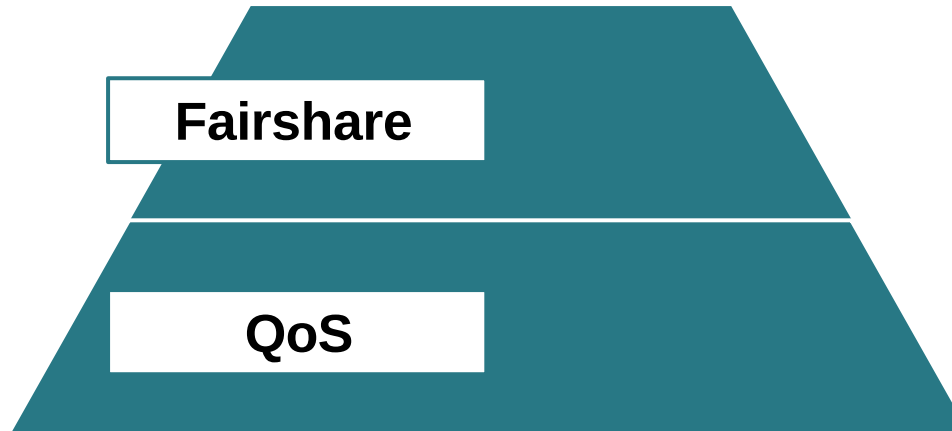
QoS

Backfill Scheduling:

- Priority factor:

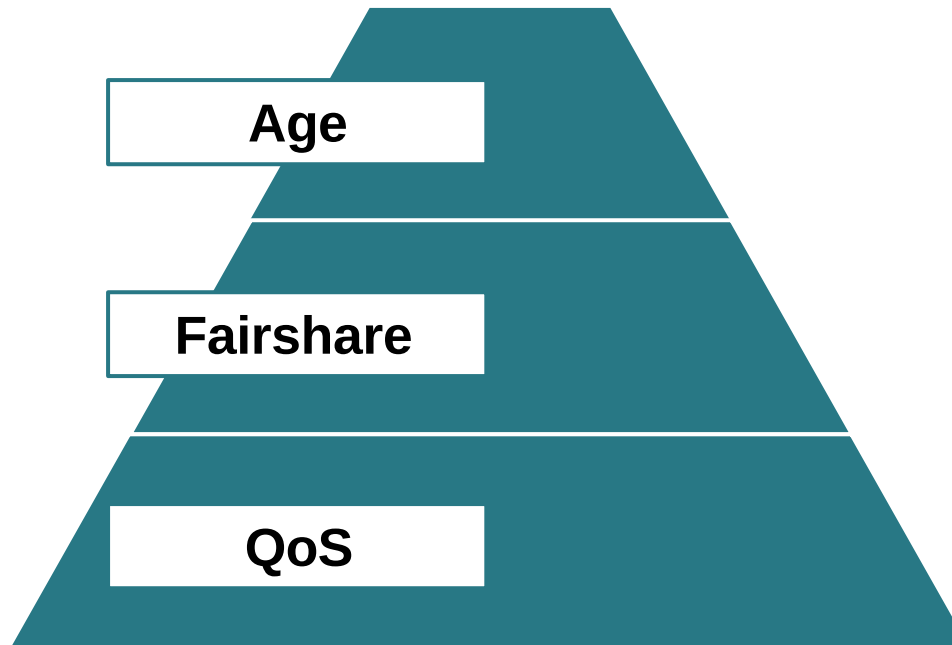
Fairshare:

- It depends on:
 - Consumption.
 - Resources requested.



Backfill Scheduling:

- Priority factor:

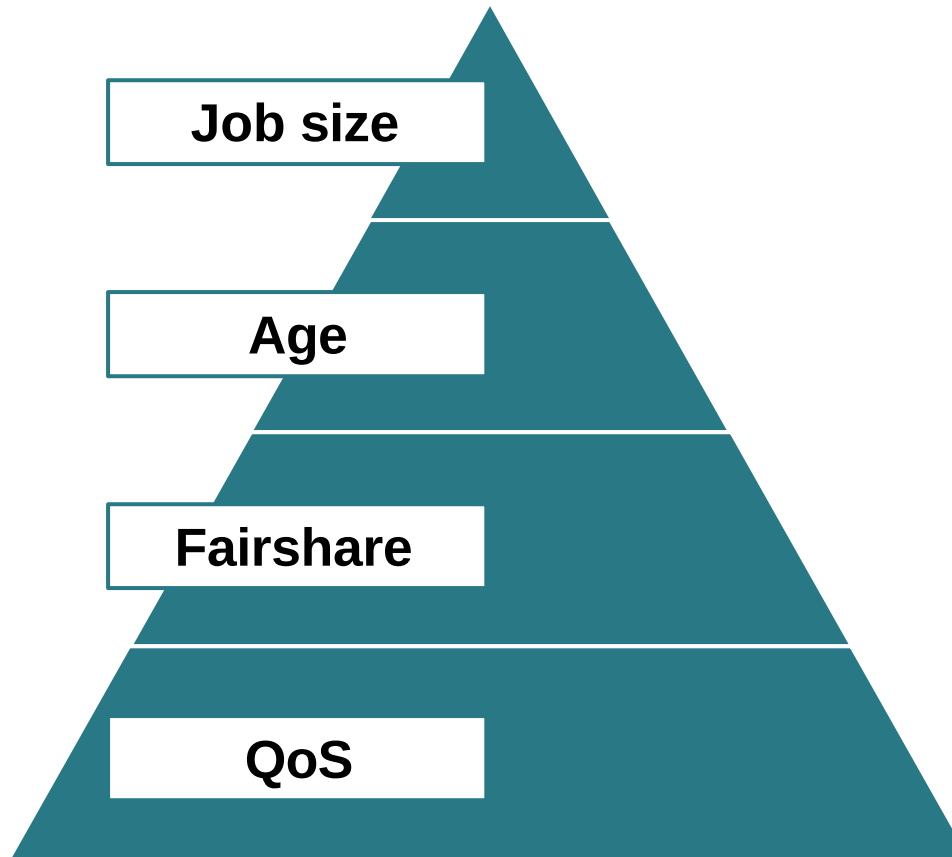


Age:

- Increase priority as more time the job pends on queue.
- For a maximum of x days.
- Not valid for dependent jobs!

Backfill Scheduling:

- Priority factor:



Job size:

- Bigger jobs have more priority.
- ONLY resources
NOT time.

Basic: Jobs



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PENDING

(CONFIGURING)
RUNNING

COMPLETING

COMPLETED



```
$ vim submit_file.slm
```

Batch job:

Submit

jobA



Execute

jobA



Obtain

jobA

```
#!/bin/bash
#SBATCH -J <job_name>
#SBATCH -o output_file.out
#SBATCH -e error_file.err
#SBATCH -p <partition>
#SBATCH -n <#tasks>
#SBATCH -c <#cpus_per_task>
#SBATCH -t 60

module load <module>

cp -r <input> ${SCRATCH}
cd ${SCRATCH}

<application>

cp -r <output> ${SLURM_SUBMIT_DIR}
```

Batch job:

Slurm directives

-J <name>	Name of job
-o <file>	Job's std output file
-e <file>	Job's std error file
-p <part>	Partition requested
-n <#tasks>	Number of tasks
-c <#cpus>	Number of procs per task
-t <time>	Time limit (dd-hh:mm, hh:mm, mm)

```
#!/bin/bash
```

```
#SBATCH -J <job_name>
```

```
#SBATCH -o output_file.out
```

```
#SBATCH -e error_file.err
```

```
#SBATCH -p <partition>
```

```
#SBATCH -n <#tasks>
```

```
#SBATCH -c <#cpus_per_task>
```

```
#SBATCH -t 60
```

```
module load <module>
```

```
cp -r <input> ${SCRATCH}
```

```
cd ${SCRATCH}
```

```
<application>
```

```
cp -r <output> ${SLURM_SUBMIT_DIR}
```


Batch job:

Defaults

- Depend on partition and QoS

```
#!/bin/bash
```

```
#SBATCH -J <job_name>
```

```
#SBATCH -o output_file.out
```

```
#SBATCH -e error_file.err
```

```
#SBATCH -p <partition>
```

```
#SBATCH -n <#tasks>
```

```
#SBATCH -c <#cpus_per_task>
```

```
#SBATCH -t 60
```

```
module load <module>
```

```
cp -r <input> ${SCRATCH}
```

```
cd ${SCRATCH}
```

```
<application>
```

```
cp -r <output> ${SLURM_SUBMIT_DIR}
```

Batch job:

- **First** Load module
- **Second** Copy inputs to SCRATCH
Change working path
- **Third** Execution
- **Forth** Get outputs back

```
#!/bin/bash
#SBATCH -J <job_name>
#SBATCH -o output_file.out
#SBATCH -e error_file.err
#SBATCH -p <partition>
#SBATCH -n <#tasks>
#SBATCH -c <#cpus_per_task>
#SBATCH -t 60
```

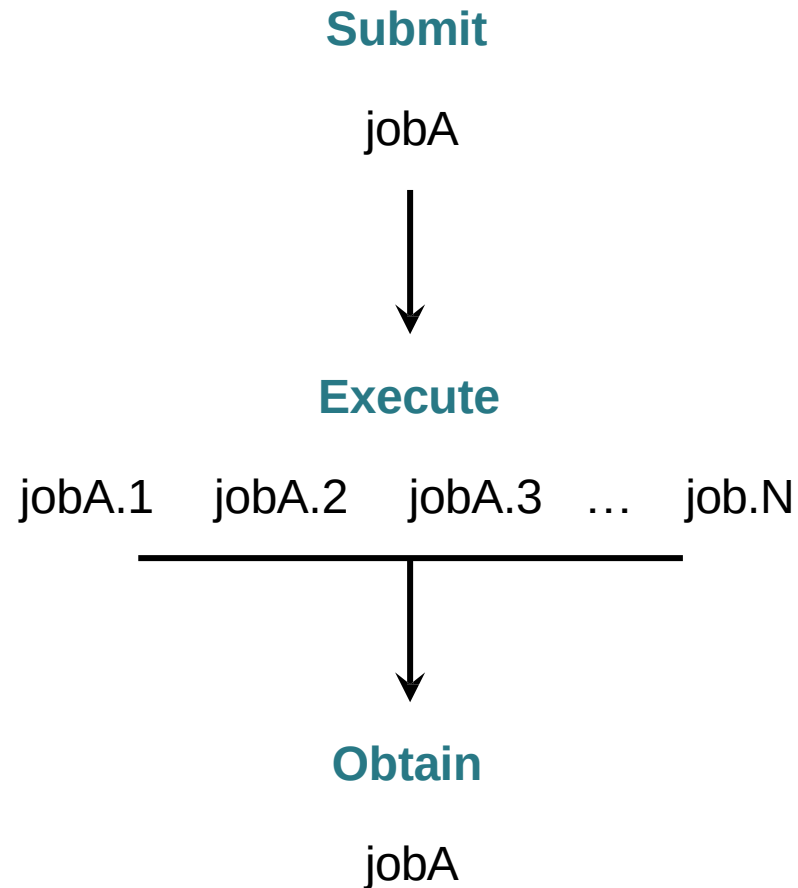
```
module load <module>

cp -r <input> ${SCRATCH}
cd ${SCRATCH}

<application>

cp -r <output> ${SLURM_SUBMIT_DIR}
```

Batch job: Steps



```
#!/bin/bash
#SBATCH -J <job_name>
#SBATCH -o output_file.out
#SBATCH -e error_file.err
#SBATCH -p <partition>
#SBATCH -n <#tasks>
#SBATCH -c <#cpus_per_task>
#SBATCH -t 60
```

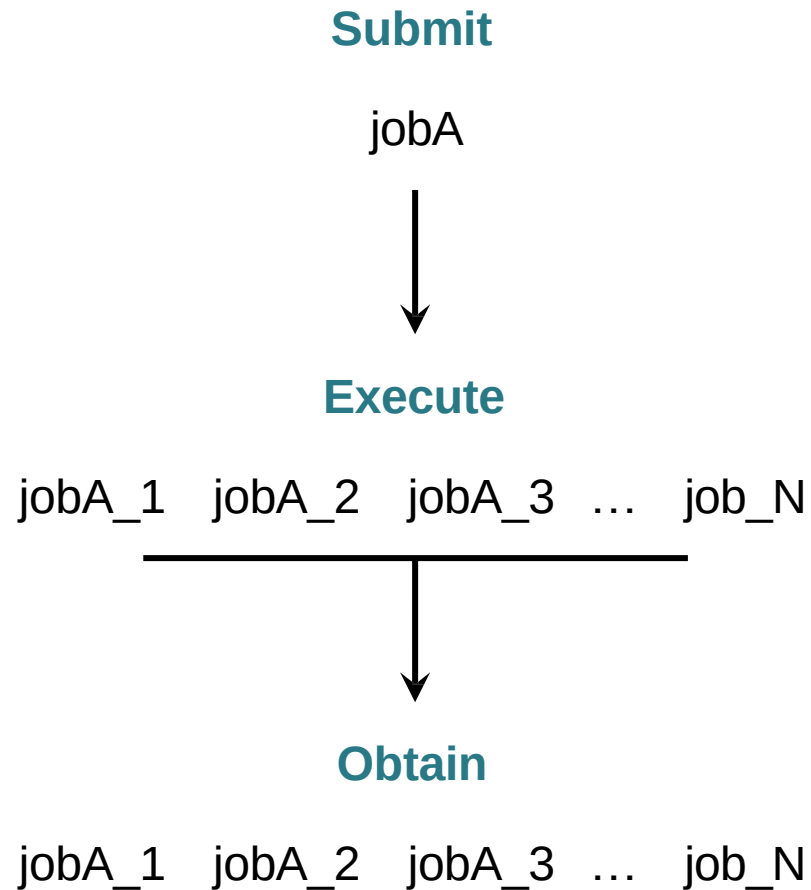
```
module load <module>
```

```
cp -r <input> ${SCRATCH}
cd ${SCRATCH}
```

```
srun <application>
srun <application> &
srun <application> &
wait
```

```
cp -r <output> ${SLURM_SUBMIT_DIR}
```

Batch job: Arrays



```
#!/bin/bash
#SBATCH -J <job name>
#SBATCH -o output_file_%A_%a.out
#SBATCH -e error_file_%A_%a.err
#SBATCH --array=0-4
#SBATCH -p <partition>
#SBATCH -n <#tasks>
#SBATCH -c <#cpus_per_task>
#SBATCH -t 60

module load <module>

cp -r <input> ${SCRATCH}
cd ${SCRATCH}

<application>

cp -r <output> ${SLURM_SUBMIT_DIR}
```

Basic: Jobs

Batch job: Arrays

- **Example: Job 50**

SLURM_JOB_ID	50
SLURM_ARRAY_JOB_ID	50
SLURM_ARRAY_TASK_ID	0
SLURM_ARRAY_TASK_COUNT	5
SLURM_ARRAY_TASK_MAX	4
SLURM_ARRAY_TASK_MIN	0
SLURM_JOB_ID	51
SLURM_ARRAY_JOB_ID	50
SLURM_ARRAY_TASK_ID	1
SLURM_ARRAY_TASK_COUNT	5
SLURM_ARRAY_TASK_MAX	4
SLURM_ARRAY_TASK_MIN	0
...	

```
#!/bin/bash
#SBATCH -J <job name>
#SBATCH -o output_file_%A_%a.out
#SBATCH -e error_file_%A_%a.err
#SBATCH --array=0-4
#SBATCH -p <partition>
#SBATCH -n <#tasks>
#SBATCH -c <#cpus_per_task>
#SBATCH -t 60

module load <module>

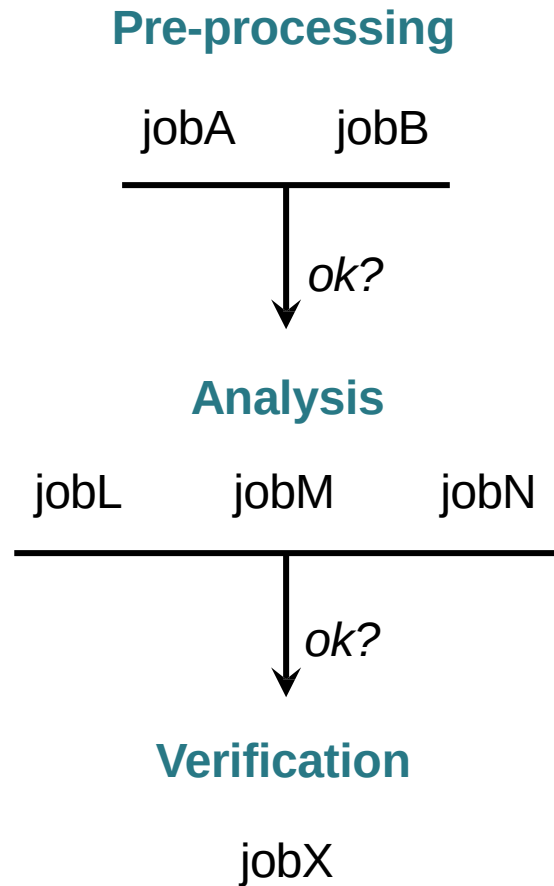
cp -r <input> ${SCRATCH}
cd ${SCRATCH}

<application>

cp -r <output> ${SLURM_SUBMIT_DIR}
```

Basic: Jobs

Batch job: Dependency



```
#!/bin/bash
#SBATCH -J <job_name>
#SBATCH -o output_file.out
#SBATCH -e error_file.err
#SBATCH -p <partition>
#SBATCH -n <#tasks>
#SBATCH -c <#cpus_per_task>
#SBATCH -t 60
#SBATCH --dependency=afterok:<jid>
```

```
module load <module>
```

```
cp -r <input> ${SCRATCH}
cd ${SCRATCH}
```

```
<application>
```

```
cp -r <output> ${SLURM_SUBMIT_DIR}
```

Basic: Jobs

Batch job: Dependency

- **after:<jobid>[:<jobid>...]**
Start of the specified jobs.
- **afterany:<jobid>[:<jobid>...]**
Termination of the specified jobs.
- **afternotok:<jobid>[:<jobid>...]**
Failing of the specified jobs.
- **singleton**
All jobs with the same name and user have ended.

```
#!/bin/bash
#SBATCH -J <job_name>
#SBATCH -o output_file.out
#SBATCH -e error_file.err
#SBATCH -p <partition>
#SBATCH -n <#tasks>
#SBATCH -c <#cpus_per_task>
#SBATCH -t 60
#SBATCH --dependency=afterok:<jid>

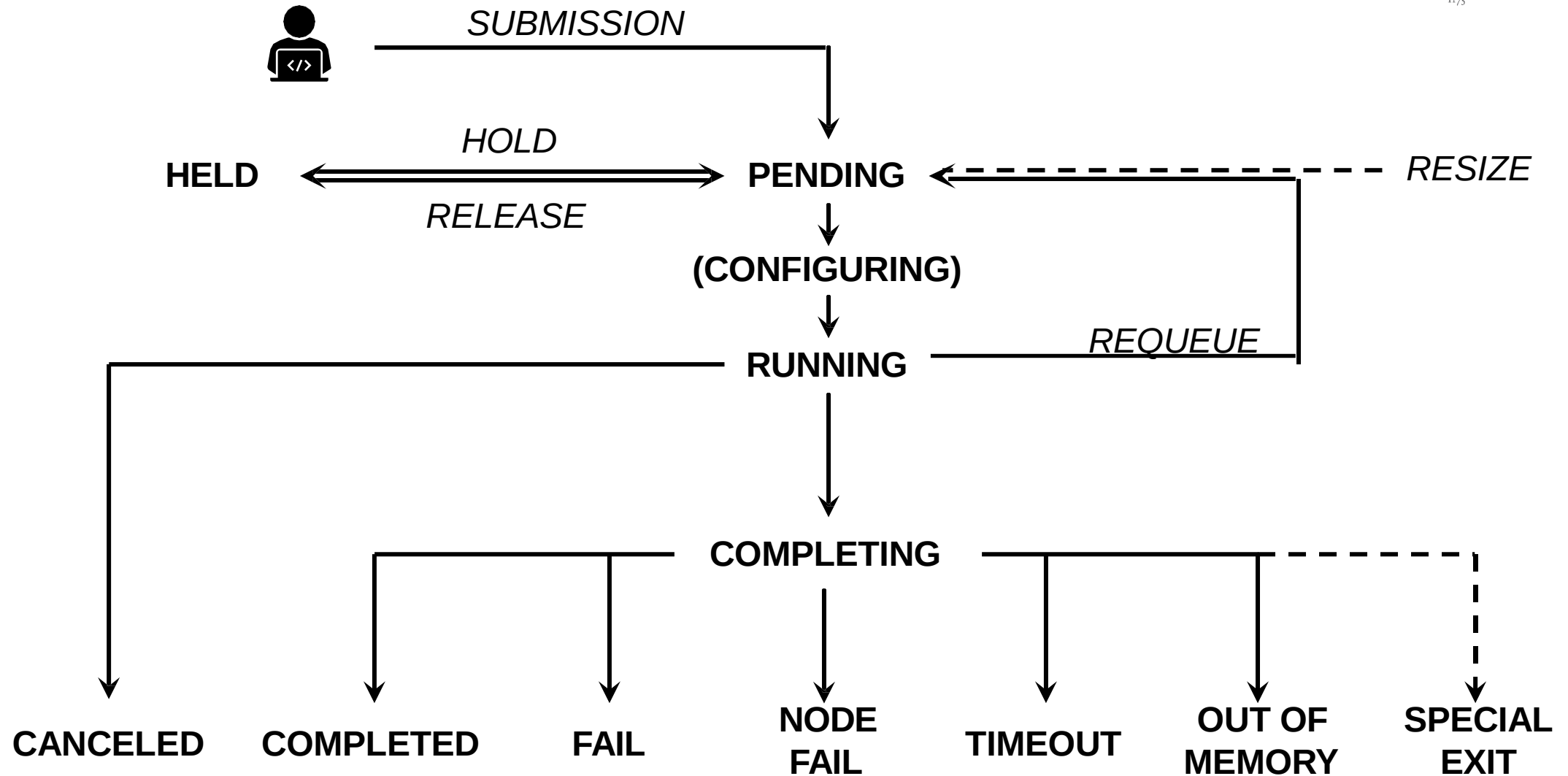
module load <module>

cp -r <input> ${SCRATCH}
cd ${SCRATCH}

<application>

cp -r <output> ${SLURM_SUBMIT_DIR}
```

Basic: Jobs





PENDING

RUNNING

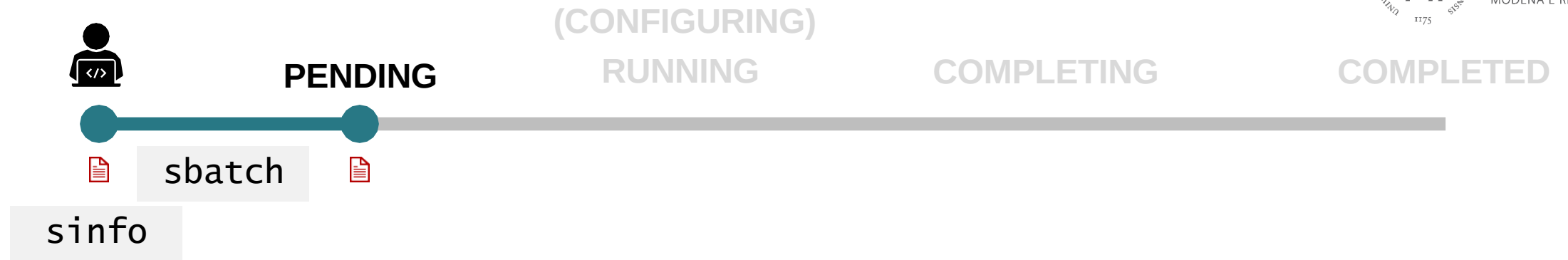
COMPLETING

COMPLETED

sinfo

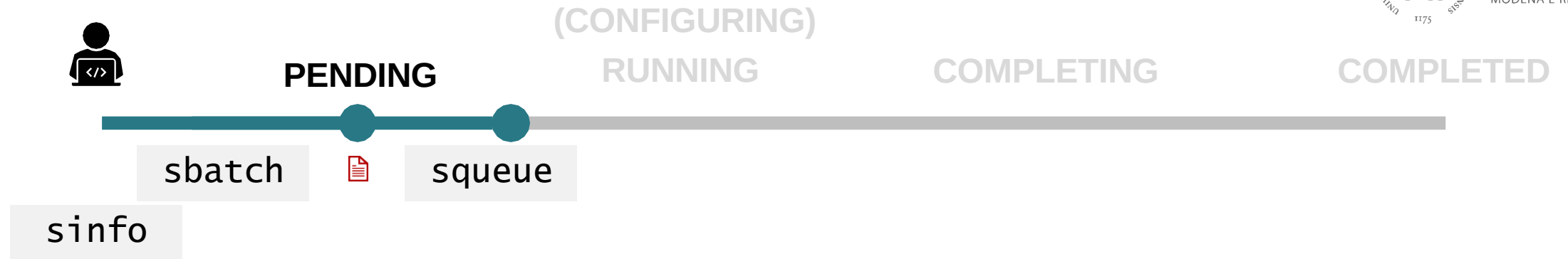
PARTITION	AVAIL	TIMELIMIT	NODES	STATE	NODELIST
boost_usr_prod	up	1-00:00:00	11	plnd	lrdn[0430,0444,0679,0698,1600,1866,2778,2819,2882,3084,3218]
boost_usr_prod	up	1-00:00:00	2	comp	lrdn[0267,3251]
boost_usr_prod	up	1-00:00:00	3	drain	lrdn[2755,2961,3041]
boost_usr_prod	up	1-00:00:00	2	resv	lrdn[0069,0241]
boost_usr_prod	up	1-00:00:00	598	mix	lrdn[0002,0004,0008-0009,0014,0016-0017,0019,0024-0026,0034-0036,0039,0041-0043,0046,0048,0050-0051,0056,0058,0061-0062,0067-0068,0070-0073,0075,0078,0083,0085-0086,0091,0094,0096,0098-0101,0106,0114,0119-0120,0126,0128,0130,0132,0134,0139,0142,0144-0145,0152-0154,0162,0164,0166,0173,0178,0180,0182-0185,0188-0189,0191,0200,0203,0211-0212,0218-0219,0222,0226,0230,0232,0234-0235,0237,0239-0240,0243,0248,0250,0252,0254-0255,0258,0260,0262-0263,0265,0268-0269,0272,0275,0283,0287,0300-0301,0304,0308-0312,0316,0321,0338-0339,0348,0351,0360,0371-0373,0376,0378-0380,0382,0386,0388-0389,0392,0395,0399-0400,0404,0418,0421-0422,0425-0426,0428-0429,0438,0442,0446-0447,0455,0467,0469,0474,0478-0479,0481,0483,0486,0488,0493,0495-0496,0513,0517-0518,0522,0528,0530,0540,0542-0543,0545,0547,0551,0557-0558,0565,0571,0579,0581,0584-0585,0591,0596-0597,0605,0607,0612,0615-0616,0633,0641,0643,0649,0657,0675,0683,0693,0699,0701,0709,0716,0724,0728-0729,0753,0757,0759,0765,0767,0770,0772,0778,0781-0783,0786,0788,0795,0802,0804,0811,0815,0825,0829,0831,0835,0850-0852,0855,0860-0861,0875,0880,0882,0887,0900,0907,0913,0920,0933,0938,0942,0975,0997,1000,1009,1018,1026,1028-1029,1032,1041,1048,1057,1063,1066,1072,1077,1080-1081,1084,1094-1095,1098-1099,1113,1118,1133,1150,1152,1175-1176,1190,1200,1217-1218,1229,1233,1235,1272-1273,1289,1292,1310,1318,1322,1330,1351,1356,1371-1372,1374,1387,1396,1411,1414,1425-1427,1432-1433,1437-1439,1441,1446,1459,1468,1525,1532-1534,1555,1564,1585,1592,1602,1605,1609,1635,1644,1651,1656,1660,1662,1668,1671,1675,1678,1683,1685,1691,1698,1702,1711-1712,1718,1720-1721,1728,1735,1747,1751,1768,1772,1778-1779,1794,1807,1811,1839,1861,1863,1871-1872,1881,1887,1892,1908,1913,1915,1918,1921-1922,1928,1931,1936,1941,1946,1949,1954-1955,1966,1969,1972,1978,1984,1987,2000,2002,2005,2010-2011,2016,2021,2042,2073,2077,2080,2098,2106,2117,2164,2186-2187,2198,2238,2245,2264,2283,2287,2295,2314-2315,2326,2338,2343-2344,2356,2373,2376,2379,2409,2416,2424,2431,2458,2461-2462,2464,2467,2481,2490,2501,2528,2532,2548,2557,2559-2560,2587,2595,2600,2609,2642,2661,2666,2670,2673,2686,2696,2701,2723,2727,2729-2730,2734,2738,2756,2758,2762,2769,2784,2796,2814,2820,2824,2830,2834,2847,2855,2860,2863,2869,2875,2877,2879,2895,2899,2902-2903,2916,2937,2939,2948,2953,2962,2997,3016,3037,3060,3086,3089,3104,3118,3120,3122,3127,3129,3133,3137,3142,3153,3155,3157,3176,3182,3184,3189,3201,3206,3211,3215-3217,3220,3225-3227,3229,3234-3235,3237,3242-3243,3245,3249,3252,3257,3260,3263,3267,3273-3274,3280,3284,3288,3291-3292,3298-3299,3301-3302,3305,3312-3313,3315,3320-3321,3326-3328,3333,3346,3348,3357-3358,3360-3363,3366,3371,3377,3381-3383,3385,3387,3392-3395,3397-3398,3400-3402,3405,3408,3410-3413,3415,3417-3422,3425-3426,3432,3436-3437,3439-3440,3444-3445,3449-3456]
boost_usr_prod	up	1-00:00:00	2669	alloc	lrdn[0001,0003,0005-0007,0010-0013,0015,0018,0020-0023,0027-0033,0037-0038,0040,0044-0045,0047,0049,0052-0055,0057,0059-0060,0063-0066,0074,0076-0077,0080-0082,0084,0087-0090,0092-0093,0097,0102-0105,0107-0113,0115-0118,0121-0125,0127,0129,0131,0133,0136-0138,0140-0141,0143,0146-0150,0155-0161,0163,0165,0167-0172,0174-0177,0181,0186-0187,0190,0192-0199,0201-0202,0204-0210,0213-0217,0220-0221,0223-0225,0227-0229,0231,0233,0236,0238,0242,0245-0247,0249,0253,0257,0259,0261,0264,0266,0270,0273-0274,0276-0282,0284-0286,0289-0294,0296-0299,0302-0303,0305-0307,0313-0315,0317,0319-0320,0322-0337,0340-0347,0349-0350,0353,0355-0359,0362,0364-0370,0375,0377,0383-0385,0387,0390-0391,0393-0394,0396-0398,0401-0403,0405-0417,0419-0420,0423-0424,0427,0431-0437,0439-0441,0443,0445,0448-0454,0457-0466,0468,0471-0473,0475-0477,0480,0482,0484-0485,0487,0489,0491-0492,0494,0497-0510,0512,0514-0516,0519-0521,0523-0527,0529,0531-0539,0541,0544,0546,0548-0550,0552-0556,0559-0562,0564,0566-0570,0572-0578,0580,0582-0583,0586-0590,0592-0595,0598-0600,0606,0608-0611,0613-0614,0617-0632,0634-0640,0642,0644-0648,0650-0656,0658-0667,0669-0670,0672-0674,0676-0678,0680-0682,0684-0685,0687-0690,0692,0694-0697,0700,0702-0708,0711-0715,0717-0723,0725-0727,0730,0732-0752,0754-0756,0758,0760-0763,0766,0768-0769,0771,0773-0777,0779-0780,0784-0785,0789-0794,0796-0797,0799-0800,0803,0805-0810,0812-0814,0816-0824,0826-0828,0832-0834,0836-0838,0840-0849,0853-0854,0856-0859,0862-0874,0876-0879,0881,0883-0884,0886,0888-0896,0898-0899,0901-0906,0908-0912,0914-0919,0921-0932,0934-0937,0939-0941,0943-0958,0960-0969,0971-0974,0977-0996,0998,1002-1003,1005-1008,1010-1012,1015-1017,1019-1023,1025,1027,1030-1031,1033-1036,1039-1040,1042-1044,1046-1047,1049-1056,1058-1062,1064-1065,1068-1071,1073-1076,1078-1079,1082-1083,1085-1093,1096-1097,1100-1108,1110-1112,1114-1115,1117,1119-1132,1134-1139,1141-1149,1151,1153,1155-1163,1165-1174,1177-1184,1186-1187,1189,1191-1199,1202-1205,1207-1208,1210-1214,1216,1219-1228,1230-1232,1234,1237-1271,1274-1288,1290-1291,1293-1309,1311-1317,1319-1321,1323-1329,1331-1350,1352-1355,1357-1370,1373,1375-1386,1388-1395,1397-1410,1412-1413,1415-1419,1421-1424,1428-1431,1434-1436,1440,1442,1444-1445,1447-1458,1460-1467,1469-1473,1475-1524,1526-1531,1535-1545,1547-1554,1556-1563,1565-1567,1569-1577,1579-1584,1586-1589,1591,1593-1599,1601,1603-1604,1606-1608,1610-1634,1636-1638,1640-1643,1645-1650,1652-1655,1657-1659,1661,1664-1667,1669-1670,1672-1674,1676-1677,1679-1682,1684,1686-1690,1692-1695,1697,1699-1701,1703-1710,1713-1717,1719,1722-1724,1726-1727,1729-1734,1736-1746,1748-1750,1753-1754,1756-1767,1769-1771,1773-1777,1780-1783,1785-1793,1795-1797,1799-1806,1808-1810,1812-1820,1822-1838,1840-1853,1855-1860,1862,1864-1865,1867-1870,1873-1880,1882-1886,1888-1891,1893-1899,1901-1907,1909-1912,1914,1916-1917,1919-1920,1923-1927,1929-1930,1932-1935,1937-1940,1942-1945,1947,1950-1953,1956-1961,1963-1965,1967-1968,1970-1971,1973-1977,1979-1983,1985-1986,1988-1999,2001,2003-2004,2006-2009,2012-2015,2017-2018,2022-2027,2029-2041,2043,2045-2066,2068-2070,2072,2074-2076,2078-2079,2081-2097,2099-2100,2102,2104-2105,2107-2116,2118-2124,2126-2147,2149,2151-2163,2165-2179,2181-2185,2188-2197,2199-2217,2219-2237,2239-2244,2246-2263,2265-2282,2284-2286,2288-2294,2296-2313,2316-2325,2327-2337,2339-2342,2345,2347-2355,2359-2362,2365-2367,2369-2372,2374-2375,2377-2378,2380-2383,2385-2387,2389,2391-2396,2398-2408,2410-2415,2417-2421,2423,2425-2430,2432-2437,2439-2457,2459-2460,2463,2465-2466,2468-2480,2482-2489,2491-2499,2502-2524,2526-2527,2529-2531,2533-2547,2549-2550,2552-2556,2558,2561,2563-2571,2573-2574,2576-2581,2584-2586,2588-2592,2594,2596-2597,2599,2601-2608,2610-2628,2630-2641,2643-2653,2655-2658,2660,2662,2664-2665,2667-2669,2671-2672,2674-2685,2687-2695,2698-2700,2702-2703,2705-2722,2724-2726,2728,2731-2733,2735-2737,2739-2754,2757,2759-2761,2763-2768,2771-2774,2776-2777,2779-2783,2785-2795,2797-2800,2802-2812,2815-2818,2821-2823,2825-2829,2831-2833,2835-2846,2848-2854,2856-2857,2859,2861-2862,2864,2866-2868,2870-2873,2876,2878,2880-2881,2883-2894,2896-2898,2900-2901,2904-2912,2914-2915,2917-2926,2928-2936,2938,2940-2947,2949-2952,2954-2960,2963-2966,2968-2974,2977-2988,2990-2992,2994-2996,2998-3011,3014-3015,3017-3023,3025-3036,3038-3040,3042-3055,3057-3059,3061-3068,3070-3079,3081-3083,3085,3087-3088,3090,3092-3098,3100-3103,3105-3110,3112-3117,3119,3121,3123-3124,3126,3128,3131-3132,3134-3136,3138-3141,3144-3150,3154,3156,3158-3167,3169-3175,3177,3179,3181,3183,3185-3186,3188,3190-3200,3202,3204-3205,3207-3208,3210,3212,3214,3219,3222-3224,3228,3230-3233,3236,3238-3241,3244,3246-3248,3250,3253-3256,3258-3259,3261-3262,3264-3266,3268-3272,3275-3278,3281-3283,3286-3287,3289-3290,3293-3297,3300,3303-3304,3306-3311,3316-3319,3322-3325,3329-3332,3334-3342,3344,3347,3349-3354,3359,3365,3367-3370,3372-3376,3378,3380,3384,3386,3388-3391,3396,3399,3403-3404,3406-3407,3409,3414,3416,3423,3427-3429,3431,3433-3435,3438,3441-3443,3446-3447]

Basic: Jobs



```
$ sbatch <file>  
Submitted batch job 1720189
```

Basic: Jobs



- **Priority:** One or more higher priority jobs exist for this partition or advanced reservation.
- **Dependency:** The job is waiting for a dependent job to complete.

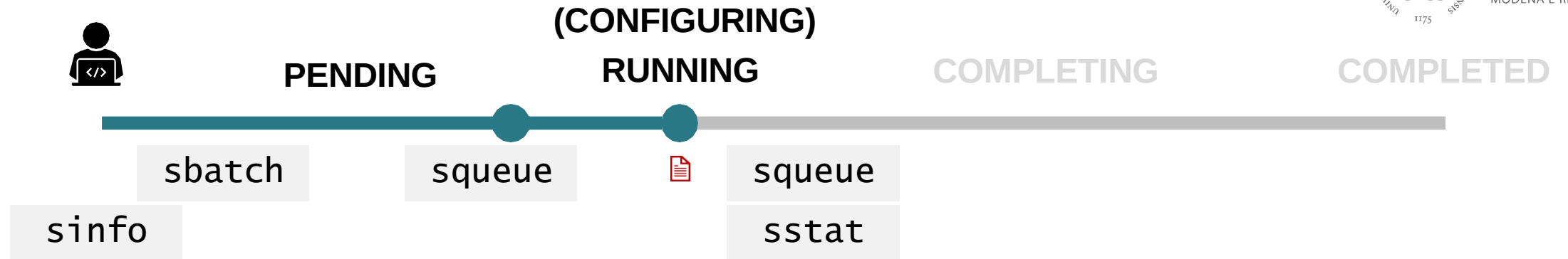
```
$ sbatch <file>
```

```
Submitted batch job 1720189
```

```
$ squeue -u <username>
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
1720189	std	test	user	PD	0:00	1	(Resources)

Basic: Jobs



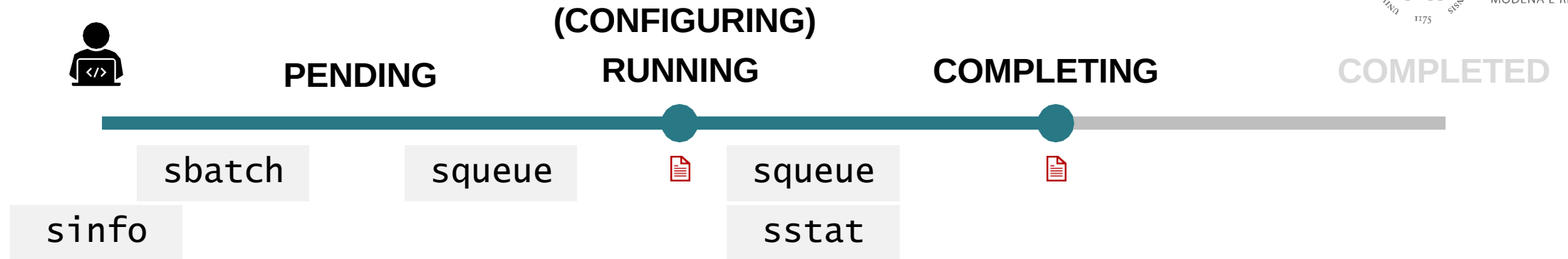
```
$ squeue -j 1720189
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
1720189	std	test	user	R	1-03:44	1	pirineus27

```
$ sstat -aj 1720189 --format=jobid,nodelist,mincpu,maxrss,pids
```

JobID	Nodelist	MinCPU	MaxRSS	Pids
1720189.ext+	pirineus27			226474
1720189.bat+	pirineus27	00:00.000	7348K	226491,226526,226528
1720189.0	pirineus27	1-03:44:05	19171808K	226557,226577

Basic: Jobs

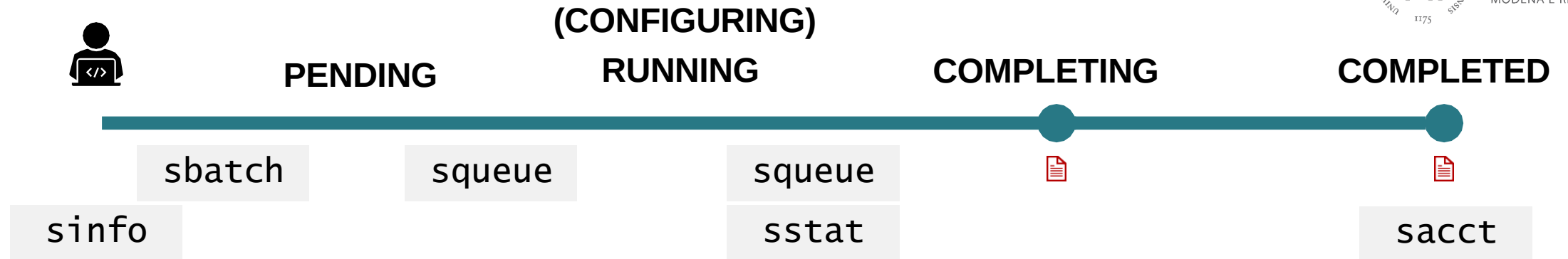


- Move files from **\$LOCALSCRATCH** to **\$SHAREDSCRATCH**.

```
$ squeue -j 1720189
```

JOBID	PARTITION	NAME	USER	ST	TIME	NODES	NODELIST(REASON)
1720189	std	test	user	CG	2-15:56	1	pirineus27

Basic: Jobs

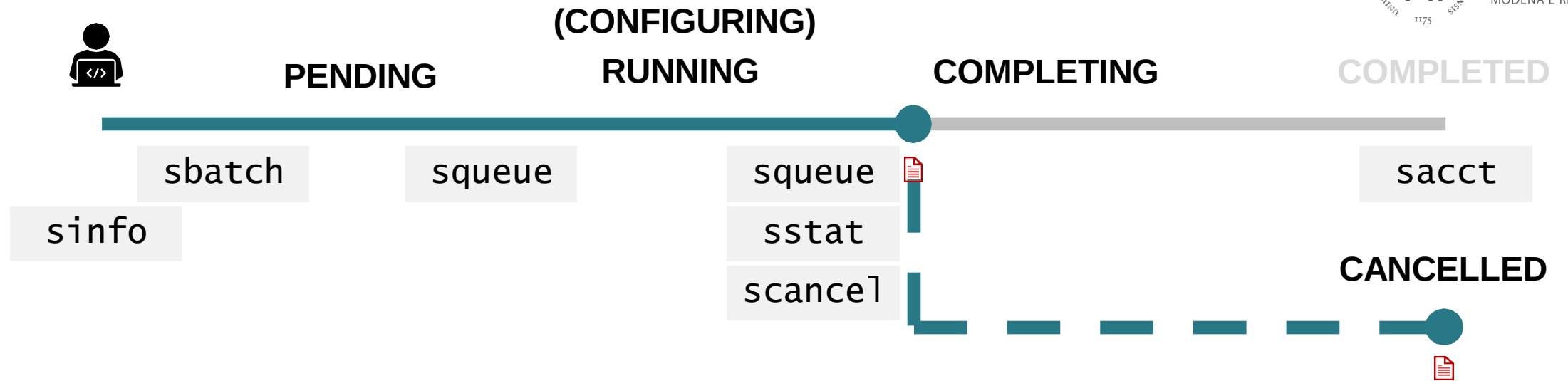


- completed (CP), time_out (TO), out of memory (OM), fail (F), node_fail (NF)...

```
$ sacct
```

JobID	JobName	Partition	Account	AllocCPUS	State	ExitCode
1720189	test	std	account	16	COMPLETED	0:0
1720189.bat+	batch		account	16	COMPLETED	0:0
1720189.ext+	extern		account	16	COMPLETED	0:0
1720189.0	pre		account	16	COMPLETED	0:0
1720189.1	process		account	16	COMPLETED	0:0
1720189.2	post		account	16	COMPLETED	0:0

Basic: Jobs



```
$ scancel -j 1720189
```

```
$ sacct
```

JobID	JobName	Partition	Account	AllocCPUS	State	ExitCode
1720189	test	std	account	16	CANCELLED+	0:0
1720189.bat+	batch		account	16	CANCELLED+	0:0
1720189.ext+	extern		account	16	COMPLETED	0:0
1720189.0	pre		account	16	CANCELLED+	0:0

IT'S
HIGHLY
ADDICTIVE



Questions?

